

synaptic: Technical Manual

Internals and Customisation

Template Development Team

CTAN Community

E-mail: synaptic@example.com

Abstract: This document explains the internal architecture of `synaptic` and provides recipes for extending its functionality: overriding labels and styles, adding theorem environments, and creating custom colour themes. It is intended for package developers and power users.

Keywords: synaptic, technical manual, internals, customisation

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To all those who craft beautiful documents with $\text{\TeX}$.

Contents

1	Architecture Overview	1
2	Customising Labels and Styles	1
3	Theorem Environment Internals	1
4	Font Loading and Fallbacks	2
5	Colour Theme Internals	2
6	Fine-Breaking Implementation	2

1 Architecture Overview

synaptic is written entirely in `expl3`. All internal macros carry the prefix `\__synaptic_` or `\g__synaptic_`. Key-value options are parsed via `l3keys2e` and stored in global variables early during package loading.

The main data structures are:

- `\g__synaptic_labels_prop` – label texts
- `\g__synaptic_styles_prop` – style macros
- `\g__synaptic_meta_prop` – title-page metadata
- `\g__synaptic_authors_seq`, `\g__synaptic_affils_seq`, `\g__synaptic_emails_seq`

All are populated via `\prop_gput:Nnn` or `\seq_gput_right:Nn`, ensuring easy extension.

2 Customising Labels and Styles

Users can override any label or style in the preamble:

```
\synaptic_register_label:nn { abstract } { Summary }
\synaptic_register_style:nn { thmhead }
{ \normalfont\bfseries\color{blue} }
```

Supported label keys: `abstract`, `keywords`, `arxiv`, `default_header`, `acknowledgments`, `proof`, `thm`, `lemma`, `prop`, `def`, `email`.

Style keys: `thmhead`, `thmnote`, `label`, `thmbody`.

Caution 2.1. Customizations should be placed before `\maketitle` to take effect.

3 Theorem Environment Internals

The predefined style `syn_thm_modern_style` is declared with `\declaretheoremstyle`. All theorem environments are derived from it and share a sibling relationship to `theorem`.

```
\declaretheorem[
  style = syn_thm_modern_style,
  name  = \synaptic_get_label:n { lemma },
  sibling = theorem
]{lemma}
```

The user-level wrapper `\declarecustomtheorem` simplifies this process.

Notation 3.1. In this document we used `notation` and `caution` as custom environments.

4 Font Loading and Fallbacks

The package checks for font availability with `\IfFontExistsTF` and provides explicit fallbacks (e.g., Latin Modern). In Chinese mode, `ctex` is loaded with `scheme=plain` to avoid clashes with `amsthm`. No unknown options are passed to `ctex`, preventing option leakage.

5 Colour Theme Internals

All colours derive from `syncolor`. Derived colours are computed via `\colorlet`:

```
\colorlet{syncolor-light}{syncolor!40}  
\colorlet{syncolor-rule}{gray!40}
```

To create a completely new theme, redefine `syncolor` and its variants in the preamble.

6 Fine-Breaking Implementation

When `fine-breaking=true`, the package sets:

```
\emergencystretch=4em  
\tolerance=4000  
\widowpenalty=10000  
\clubpenalty=10000  
\raggedbottom
```

These parameters reduce overfull boxes and widow/orphan lines without drastically altering the overall look.

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